

Calculations

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1. Delta Estimation

This is supposed to be a back of the envelope calculation to determine how the error between a Continuous CA with Moore Neighbourhood (hereby named Vanilla) and a Continuous CA with a distant neighbour (hereby referred to as Small World) will grow, after each step of the evolution.

Assume a grid of $N \times N$ cells, and the state of a cell represented as X_{ij}^t . For a vanilla, the state at time $t + 1$ can be calculated as

$${}_L X_{ij}^{t+1} = f_L \left(X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t \right) \quad (1)$$

For a small world model, if we assume that every cell gets a partner outside its Moore neighbourhood, the evolution equation is of the form

$${}_{NL} X_{ij}^{t+1} = f_L \left(X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t, X_{i+n,j+m}^t \right) \quad (2)$$

where $X_{i+k,j+l}^t$ is the long distance neighbour and $X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t$ represents the Moore neighbourhood. Usually, $X_{i+n,j+m}^t$ is independent of $X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t$.

But, we make the assumption that a good enough f_L can approximate $X_{i+n,j+m}^t$ with some error ε .

$$X_{i+k,j+l}^t = f_A \left(X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t \right) + \varepsilon$$

which means that with some $f_L = f_{AL}$

$$\begin{aligned} {}_{NL} X_{ij}^{t+1} &= f_{AL} \left(X_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^t \right) + \varepsilon \\ &= {}_L \hat{X}_{ij}^{t+1} + \varepsilon \end{aligned} \quad (3)$$

At time $= t + 1$, the delta between the approximation and the small world cell state is

$$\Delta(t + 1) = \left| {}_{NL} X_{ij}^{t+1} - {}_L \hat{X}_{ij}^{t+1} \right| = \varepsilon \quad (4)$$

At time $= t + 2$,

$${}_L \hat{X}_{ij}^{t+2} = f_{AL} \left(\hat{X}_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^{t+1} \right) + \varepsilon \quad (5)$$

$\hat{X}_{\{i-1,i,i+1\}\{j-1,j,j+1\}}^{t+1}$ is composed of 9 terms, and each term has an ε term associated with it. Therefore,

$$\Delta(t + 2) = \sum_{\text{Moore}} \left| \frac{\partial f}{\partial X_{ij}} \right| \left| \hat{X}_{ij}^{t+1} - X_{ij}^{t+1} \right| + \varepsilon \quad (6)$$

If we consider the leading $\left| \frac{\partial f}{\partial X_{ij}} \right|$ to be α , the equation reduces to

$$\Delta(t + 2) \leq (9\alpha + 1)\varepsilon \quad (7)$$

Similarly, for $t + 3$,

$$\begin{aligned} \Delta(t + 3) &\leq 9\alpha\Delta(t + 2) + \varepsilon \\ &\leq 81\alpha^2\varepsilon + 9\alpha\varepsilon + \varepsilon \\ &= (81\alpha^2 + 9\alpha + 1)\varepsilon \end{aligned} \quad (8)$$

More generally,

$$\begin{aligned} \Delta(t + k) &\leq \sum_{0 \leq j < k} (9\alpha)^j \varepsilon \\ &= \frac{(9\alpha)^k - 1}{9\alpha - 1} \varepsilon \end{aligned} \quad (9)$$

If $\alpha < \frac{1}{9}$, as $k \rightarrow \infty$,

$$\Delta(t + k) \leq \frac{1}{1 - 9\alpha} \varepsilon \quad (10)$$

1.1. Incorporating Time

CAs are Turing complete. This means that given enough time, it can compute any function, which implies there are no functions which the small world can compute that the vanilla cannot, **given enough time**. So the difference in capabilities can only be measured given a constraint in time.

Another point to note is that, ε is not independent of time, and information from a far away point can reach a cell via diffusion, which would bring down it's ε in determining the state of $X_{i+n,j+m}$.

This would warrant certain changes in the calculations done in Section 1.

Equation 8 would have to be re written as

$$\begin{aligned} \Delta(t + 3) &\leq 9\alpha\Delta(t + 2) + \varepsilon(3) \\ &\leq 81\alpha^2\varepsilon(1) + 9\alpha\varepsilon(2) + \varepsilon(3) \end{aligned} \quad (11)$$

Equation 9 would take the form

$$\Delta(t + k) = \sum_{0 \leq j < k} (9\alpha)^j \varepsilon(k - j) \quad (12)$$

Now how would ε behave as the grid evolves?